

STANDARD OPERATING PROCEDURE

Maize & Wheat Redox Phenotyping

Bench protocols for ROS localisation, H_2O_2 , oxidative damage, antioxidant buffering and recovery

Species: *Zea mays* L. (maize) and *Triticum aestivum* L. (wheat) Assay index

Tier	Protocol	Practical reference
Localisation	P1 DAB (H_2O_2)	Daudi & O'Brien 2012
	P2 NBT ($O_2^{\bullet-}$)	Kumar et al. 2014
	P3 H_2DCFDA (general ROS)	Kaur et al. 2016
Quantification	P4 Tissue H_2O_2 , Amplex Red (preferred)	Brumbarova et al. 2016
	P5 Tissue H_2O_2 , KI (screen)	Junglee et al. 2014
	P6 Root H_2O_2 release	Liszkay et al. 2004
Damage	P7 TBARS/MDA	Hodges et al. 1999
Buffering	P8 Enzyme extraction + protein	Noctor et al. 2016
	P9 SOD	Giannopolitis & Ries 1977
	P10 CAT	Aebi 1984
	P11 APX	Nakano & Asada 1981
	P12 GR	Foyer & Halliwell 1976
	P13 Glutathione (GSH/GSSG)	Queval & Noctor 2007
	P14 Ascorbate (ASC/DHA)	Noctor & Mhamdi 2022
Advanced	A1–A5 (XTT, RBOH, $CeCl_3$, EPR, roGFP)	See Part C

Part A – Sampling rules

- **Replication:** at least **5 biological replicates** per genotype × treatment × time point. A pool of 2–3 plants counts as one replicate. **Average technical duplicates before statistics.**
- **Time points:** T0 baseline · T2 peak stress · T3 early recovery · T4 late recovery (T1 early stress is optional). Harvest time-matched controls at **every** time point.
- **Time of day:** harvest within the same 1–2 h window, interleaving treatments (control R1 → flood R1 → ...).
- **Fresh stream** (P1–P3, P6, A3): process immediately, never freeze. **Frozen stream** (P4, P5, P7–P14): liquid N₂ within 10 s, grind without thawing, store at –80 °C, assay within about **2 weeks (no more)**.
- **Normalisation:** weigh FW immediately and keep a parallel subsample for DW. Report both.

Tissue	Wheat	Maize
Leaf	Youngest fully expanded leaf, mid-lamina (flag leaf at anthesis)	Youngest fully expanded leaf, mid-lamina between midrib and margin; record leaf number
Root	Seminal roots, 0–1 and 1–3 cm from the apex; nodal roots later (never mix root types)	Primary root, 0–3 / 3–7 / 7–12 mm zones; defined nodal whorl from about V6
Watch for	Wax (add surfactant for infiltration); anthocyanins (use blanks)	Thick lamina; sugars inflate TBARS (use the P7 correction)

Flooded plants. Re-exposing hypoxic tissue to air causes a reoxygenation ROS burst (Pucciariello & Perata 2021). Fix and record the interval from de-submergence to liquid N₂ (target ≤ 30 s), and apply the same handling to controls.

Frozen powder per replicate	Amount
P4 H ₂ O ₂ (Amplex)	30–50 mg (+ a spike aliquot for QC samples)
P7 TBARS	100 mg
P8–P12 enzymes + protein (general extract + APX extract)	2 × 100 mg
P13–P14 glutathione + ascorbate (one HCl extract)	50–100 mg

Part B – Bench protocols

P1. DAB staining – H₂O₂ localisation

Practical reference,

Daudi, A., & O'Brien, J. A. (2012). Detection of Hydrogen Peroxide by DAB Staining in Arabidopsis Leaves. *Bio-protocol*, 2(18), e263. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4932902/> or <https://bio-protocol.org/epdf/263>

Use: Where H₂O₂ accumulates (semi-quantitative). Output: % stained area and/or DAB optical density.

Reagents

- DAB stain: 1 mg mL⁻¹ DAB dissolved at pH 3.0 (HCl), then 0.05% (v/v) Tween-20 and 10 mM Na₂HPO₄ added. Make fresh and keep dark. DAB is a suspected carcinogen: wear gloves.
- Bleaching solution: ethanol : acetic acid : glycerol = 3 : 1 : 1.

Procedure

1. Cut standardised pieces: wheat 3–4 cm leaf segments; maize 1–1.5 cm lamina pieces avoiding the midrib; roots by defined zone.
2. Immerse in DAB stain (≥ 1 mL per sample) and vacuum-infiltrate for 5 min.
3. Incubate 4–5 h in the dark on a shaker at 80–100 rpm.
4. Leaves: bleach at 90–95 °C for 15 ± 5 min and repeat with fresh solution until clear (maize needs longer). Roots: no bleaching needed.
5. Image with fixed light, exposure and a scale bar. Fiji: Colour Deconvolution ('H DAB') → DAB channel → one fixed threshold → % area.

Controls: Pre-infiltrate with 10 mM ascorbate or catalase (negative); add exogenous H₂O₂ (positive); an unstained bleached sample (background). Score images blind.

P2. NBT staining – superoxide localisation

Practical reference,

Kumar, D., Yusuf, M. A., Singh, P., Sardar, M., & Sarin, N. B. (2014). Histochemical Detection of Superoxide and H₂O₂ Accumulation in Brassica juncea Seedlings. *Bio-protocol*, 4(8), e1108. <https://bio-protocol.org/epdf/1108>

Use: Where O₂•⁻ is produced (semi-quantitative). Output: % formazan area; report the SOD-sensitive fraction.

Reagents

- NBT stain: 0.2% (w/v) NBT in 50 mM sodium phosphate, pH 7.5. Keep dark.
- Specificity control: stain plus SOD (e.g., 100 U mL⁻¹, **pilot**).
- Bleaching: absolute ethanol in a boiling water bath.

Procedure

1. Cut tissue as in P1 and immerse in NBT stain (± SOD).
2. Vacuum-infiltrate with the same settings as P1.
3. Incubate in the dark. The reference uses overnight incubation for whole seedlings; for infiltrated leaf segments or root tips use 1–4 h (**pilot**).

4. Bleach in ethanol at about 100 °C for about 10 min, longer if needed.
5. Image and threshold the blue formazan with fixed settings.

Calculation SOD-sensitive signal = signal(−SOD) − signal(+SOD).

Controls: SOD co-incubation in every run; an unstained bleached sample.

Note NBT can also be reduced by other reductants. Do not report absolute superoxide concentrations.

P3. H₂DCFDA fluorescence – general ROS (roots)

Practical reference,

Kaur, N., Sharma, I., Kirat, K., & Pati, P. K. (2016). Detection of Reactive Oxygen Species in *Oryza sativa* L. (Rice). *Bio-protocol*, 6(24), e2061. www.bio-protocol.org/e2061

Use: Relative spatial index of oxidative activity (not specific to H₂O₂). Output: background-corrected mean fluorescence per zone.

Reagents

- 10 mM H₂DCFDA stock in DMSO (store at −20 °C, dark).
- 10 μM H₂DCFDA working solution in buffer (e.g., 10 mM MES-KCl pH 6.1 or 10 mM Tris-HCl pH 7.2, **pilot**); final DMSO ≤ 0.1%.

Procedure

1. Excise root tips of the defined zone.
2. Vacuum-infiltrate in 10 μM H₂DCFDA for 5 min, then incubate 10 min in the dark (the reference timings).
3. Wash 3 times in buffer.
4. Image within 10 min at excitation 488 nm, emission 510–530 nm, with laser power and gain fixed for all samples.
5. Measure ROIs at fixed distances from the apex (wheat 0–1 and 1–3 mm; maize 0–3 and 3–7 mm).

Controls: Unstained tissue (autofluorescence); DMSO only; exogenous H₂O₂ (positive).

Note DCF photo-oxidises and depends on esterase activity and probe uptake, which can differ between root zones. Use only for relative comparisons.

P4. Tissue H₂O₂ – Amplex Red/HRP (preferred quantitative assay)

Practical reference,

Brumbarova, T., Le, C. T. T., & Bauer, P. (2016). Hydrogen Peroxide Measurement in Arabidopsis Root Tissue Using Amplex Red. *Bio-protocol*, 6(21), e1999. <https://bio-protocol.org/en/bpdetail?id=1999&type=0>

Use: H₂O₂ content of leaf or root. Output: nmol H₂O₂ g^{−1} FW (and DW).

Reagents

- Extraction buffer: 20 mM K₂HPO₄, pH 6.5, ice-cold.
- Working solution (make fresh, keep dark): 100 μM Amplex Red + 0.2 U mL^{−1} HRP in buffer.

- H_2O_2 standards 0–10 μM in extraction buffer (0–5 μM final). Check the stock at A240, $\epsilon = 43.6 \text{ M}^{-1} \text{ cm}^{-1}$.
- Catalase, e.g., 1,000 U mL^{-1} (for the specificity control).

Procedure

1. Weigh about 30 mg frozen powder into a pre-cooled tube and record the exact mass.
2. Add 200 μL ice-cold extraction buffer (keep this ratio if scaling up) and mix by flipping until homogeneous.
3. Centrifuge 3 min at $16,200 \times g$, 4 $^\circ\text{C}$, and move the supernatant to a cold tube. Work quickly on ice.
4. Split the extract: aliquot A untreated; aliquot B + catalase (about 10 U, 10 min). Add an equal volume of buffer to A.
5. In a black plate, mix 25 μL sample or standard with 25 μL working solution, in technical duplicate.
6. Incubate 30 min at room temperature in the dark.
7. Read fluorescence at excitation 530–560 nm, emission 590 nm.

Calculation C_{sample} (μM) from the standard curve, using the A – B difference (the catalase-sensitive signal).

H_2O_2 ($\text{nmol g}^{-1} \text{ FW}$) = $C_{\text{sample}} \times V_{\text{buffer}} (\text{mL}) / m (\text{g}) \times \text{DF}$ (DF = any dilution, e.g., the catalase split).

Controls: Catalase aliquot for every sample; reagent-only wells; a pooled QC extract on every plate. **Spike recovery** per treatment matrix: add a known amount of H_2O_2 to a second portion of powder; accept 80–120%.

Note Chakraborty et al. (2016) found that perchloric acid, TCA, PVPP and activated charcoal degraded H_2O_2 or interfered with the Amplex Red assay. Extract in phosphate buffer as above, and rely on the catalase control and spike recovery for specificity.

P5. Tissue H_2O_2 – KI method (screening)

Practical reference Junglee, S., Urban, L., Sallanon, H., & Lopez-Lauri, F. (2014). Optimized Assay for Hydrogen Peroxide Determination in Plant Tissue Using Potassium Iodide. *American Journal of Analytical Chemistry*, 5, 730–736.

Use: Fast comparative screen between treatments. Use P4 for publication-grade absolute values.

Reagents

- One-step reagent per sample: 0.25 mL 0.1% (w/v) TCA + 0.5 mL 1 M KI + 0.25 mL 10 mM potassium phosphate. The reference optimised pH per tissue (pH 5.8 for leaves), so **pilot** it.
- Blank reagent: the same mixture with water instead of KI.
- H_2O_2 standards prepared in 0.1% TCA.

Procedure

1. Homogenise 150 mg frozen powder directly in 1 mL one-step reagent. Prepare a parallel blank with the blank reagent.
2. Incubate 10 min at 4 $^\circ\text{C}$, protected from light.
3. Centrifuge 15 min at $12,000 \times g$, 4 $^\circ\text{C}$.
4. Transfer 200 μL supernatant to a UV-transparent plate and incubate 20 min at 20–22 $^\circ\text{C}$ in the dark.

5. Read A350 and subtract the matching blank.

Calculation H_2O_2 ($\mu\text{mol g}^{-1}$ FW) from the TCA-matrix standard curve $\times 1 \text{ mL} / \text{m (g)}$.

Controls: A per-sample no-KI blank is mandatory, because pigments, especially wheat anthocyanins, otherwise inflate values.

P6. Apoplastic H_2O_2 release from living roots

Practical reference

Liszskay, A., van der Zalm, E., & Schopfer, P. (2004). Production of Reactive Oxygen Intermediates ($\text{O}_2^{\cdot-}$, H_2O_2 , and $\cdot\text{OH}$) by Maize Roots and Their Role in Wall Loosening and Elongation Growth. *Plant Physiology*, 136, 3114–3123. <https://pubmed.ncbi.nlm.nih.gov/15466236/>
& <https://link.springer.com/article/10.1007/s00425-003-1028-1>

Use: Non-destructive rate of H_2O_2 release by root segments. Output: $\text{nmol g}^{-1} \text{FW h}^{-1}$.

Reagents

- Rinse medium: 0.5 mM CaCl_2 + 5 mM MES, pH 6.0.
- Assay medium: rinse medium + 50 μM Amplex Red + 0.1 U mL^{-1} HRP (dark).

Procedure

1. Excise root segments of the defined zone and rinse 30 min in rinse medium to let the wound burst subside.
2. Blot, weigh, and place (e.g., 10 segments) in 1 mL assay medium in a 24-well plate (**pilot**).
3. At 15, 30 and 60 min, transfer 100 μL medium to a black plate and read immediately (excitation 530–560 nm, emission 590 nm).
4. Quantify against standards made in assay medium.

Calculation Release = slope of H_2O_2 (nmol) against time (h) over the linear phase / root FW (g).

Controls: Medium without roots (probe auto-oxidation); roots + catalase (H_2O_2 specificity); optional 10–20 μM DPI (not specific to RBOH).

P7. TBARS/MDA – interference-corrected

Practical reference Hodges, D. M., DeLong, J. M., Forney, C. F., & Prange, R. K. (1999). Improving the thiobarbituric acid-reactive-substances assay for estimating lipid peroxidation in plant tissues containing anthocyanin and other interfering compounds. *Planta*, 207, 604–611.

Use: Membrane lipid peroxidation. Output: $\text{nmol MDA equivalents g}^{-1} \text{FW}$.

Reagents

- 80% (v/v) ethanol.
- +TBA: 20% TCA, 0.01% BHT, 0.65% TBA. –TBA: 20% TCA, 0.01% BHT.

Procedure

1. Homogenise 100 mg powder in 1 mL 80% ethanol and centrifuge 10 min at $16,000 \times g$, 4 °C.

2. Mix 0.5 mL supernatant with 0.5 mL +TBA; mix a second 0.5 mL with 0.5 mL –TBA.
3. Heat 25 min at 95 °C, cool on ice, and centrifuge.
4. Read A440, A532 and A600 of both tubes.

Calculation $A = (A532 - A600)$ of +TBA – $(A532 - A600)$ of –TBA; $B = (A440 - A600)$ of +TBA × 0.0571

MDA (nmol mL⁻¹) = $[(A - B) / 157,000] \times 10^6$. Following Landi (2017), ϵ applies to $(A - B)$.

nmol g⁻¹ FW = MDA × 2 (1:1 mix) × 1 mL / m (g). Assumes a 1-cm path; correct if using a plate reader.

Controls: Reagent blanks for +TBA and –TBA; a pooled QC extract.

P8. Enzyme extraction and soluble protein

Practical reference,

Noctor, G., Mhamdi, A., & Foyer, C. H. (2016). Oxidative stress and antioxidative systems: recipes for successful data collection and interpretation. *Plant, Cell & Environment*, 39, 1140–1160
<https://pubmed.ncbi.nlm.nih.gov/26864619/> (see the method boxes).

Use: Extracts for P9–P12 and a protein value for normalisation.

Reagents

- General extract (SOD, CAT, GR): 50 mM K-phosphate pH 7.0, 1 mM EDTA, 2% (w/v) PVPP, 0.1% Triton X-100 (optional 1 mM PMSF).
- APX extract: the same buffer + 2 mM ascorbate. Keep it separate, because ascorbate interferes with the NBT-based SOD assay.
- Bradford reagent; BSA standards made up **in extraction buffer**, because Triton X-100 affects the dye response.

Procedure

1. Add 1 mL ice-cold buffer to 100 mg powder (one tube per buffer).
2. Homogenise on ice; centrifuge 15 min at 16,000 × g, 4 °C; keep the supernatant on ice.
3. Measure protein (Bradford) on the same extract.
4. Assay all enzymes the same day, and APX within about 2 h.

Calculation Activity ($\mu\text{mol min}^{-1} \text{mg}^{-1} \text{protein}$) = $(\Delta A \text{ min}^{-1} \times V_{\text{reaction}}) / (\epsilon \times l \times V_{\text{extract}} \times \text{protein})$
 ϵ in $\text{mM}^{-1} \text{cm}^{-1}$; l = path length in cm (correct it in plates); volumes in mL; protein in mg mL^{-1} . Also report per g FW.

Controls: Assay-specific blanks (see P9–P12); a pooled QC extract on every plate.

P9. SOD

Practical reference,

Giannopolitis, C. N., & Ries, S. K. (1977). Superoxide Dismutases: I. Occurrence in Higher Plants. *Plant Physiology*, 59, 309–314. <https://pubmed.ncbi.nlm.nih.gov/16659839/>

Use: Total SOD activity (U mg^{-1} protein).

Reagents

- Final: 50 mM K-phosphate pH 7.8, 13 mM methionine, 75 μM NBT, 0.1 mM EDTA, 2 μM riboflavin (added last), extract.

Procedure

1. Set up at least 3 extract volumes per sample, plus a no-extract tube (maximal colour) and a dark tube (blank).
2. Add riboflavin last and illuminate 15 min under uniform fluorescent light.
3. Read A560.

Calculation % inhibition = $(A_{\text{no extract}} - A_{\text{sample}}) / A_{\text{no extract}} \times 100$. 1 U = the amount giving 50% inhibition (interpolate from the volume series).

Controls: Dark blank; illuminated no-extract reference.

P10. CAT**Practical reference,**

Aebi, H. (1984). Catalase in vitro. *Methods in Enzymology*, 105, 121–126. 10.1016/S0076-6879(84)05016-3

Use: Catalase activity ($\mu\text{mol H}_2\text{O}_2 \text{ min}^{-1} \text{ mg}^{-1} \text{ protein}$).

Reagents

- 50 mM K-phosphate pH 7.0 + 10 mM H_2O_2 ; quartz cuvette or UV-transparent plate.

Procedure

1. Start the reaction with extract and read A240 every 5–10 s for 1 min.
2. Use only the initial linear rate (first 30–60 s), because catalase is inactivated by its substrate.

Calculation $\epsilon_{240} = 0.0436 \text{ mM}^{-1} \text{ cm}^{-1}$ ($43.6 \text{ M}^{-1} \text{ cm}^{-1}$).

Controls: No-extract (spontaneous H_2O_2 decomposition); extract without H_2O_2 (background absorbance).

P11. APX**Practical reference,**

Nakano, Y., & Asada, K. (1981). Hydrogen Peroxide is Scavenged by Ascorbate-specific Peroxidase in Spinach Chloroplasts. *Plant and Cell Physiology*, 22, 867–880.
<https://doi.org/10.1093/oxfordjournals.pcp.a076232>

Use: APX activity ($\mu\text{mol ascorbate min}^{-1} \text{ mg}^{-1} \text{ protein}$). Use the APX extract.

Reagents

- 50 mM K-phosphate pH 7.0, 0.5 mM ascorbate, 0.1 mM EDTA, extract; start the reaction with 0.1 mM H_2O_2 .

Procedure

1. Record a baseline, add H_2O_2 , then read the decrease in A290 for 1 min.

Calculation $\epsilon_{290} = 2.8 \text{ mM}^{-1} \text{ cm}^{-1}$.

Controls: The rate without H_2O_2 (non-peroxidase ascorbate oxidation) is subtracted.

P12. GR**Practical reference,**

Foyer, C. H., & Halliwell, B. (1976). The presence of glutathione and glutathione reductase in chloroplasts: A proposed role in ascorbic acid metabolism. *Planta*, 133, 21–25.

<https://link.springer.com/article/10.1007/BF00386001>

Use: GR activity ($\mu\text{mol NADPH min}^{-1} \text{mg}^{-1} \text{protein}$).

Reagents

- 100 mM K-phosphate pH 7.5, 1 mM EDTA, 0.2 mM NADPH, extract; start the reaction with 0.5 mM GSSG.

Procedure

1. Record a 1-min baseline, add GSSG, then read the decrease in A340 for 2–3 min.

Calculation $\epsilon_{340}(\text{NADPH}) = 6.22 \text{ mM}^{-1} \text{ cm}^{-1}$.

Controls: The GSSG-independent baseline rate is subtracted.

P13. Glutathione (GSH/GSSG)**Practical reference,**

Queval, G., & Noctor, G. (2007). A plate reader method for the measurement of NAD, NADP, glutathione, and ascorbate in tissue extracts: Application to redox profiling during Arabidopsis rosette development. *Analytical Biochemistry*, 363, 58–69. [10.1016/j.ab.2007.01.005](https://doi.org/10.1016/j.ab.2007.01.005)

Use: Glutathione pool size and % reduced (buffering capacity).

Reagents

- 0.2 M HCl (extraction); 0.2 M NaOH plus phosphate buffer for neutralisation to pH 5–6.
- Recycling mix: phosphate/EDTA buffer, NADPH, DTNB, glutathione reductase (concentrations as in the reference).
- 2-Vinylpyridine (2-VP; toxic, use a fume hood); GSH and GSSG standards.

Procedure

1. Extract 50–100 mg powder in 1 mL ice-cold 0.2 M HCl and centrifuge 10 min at $16,000 \times g$, 4 °C.
2. Neutralise an aliquot to pH 5–6 on ice. Keep the rest for P14.
3. **Total:** GR–DTNB recycling; read the A412 rate against GSH standards.
4. **GSSG:** treat a second aliquot with 2-VP (e.g., 1 μL per 200 μL , 30 min at room temperature), then assay against 2-VP-treated GSSG standards.

Calculation In GSH equivalents: **GSH = total – 2 × GSSG**; % reduced = $\text{GSH} / \text{total} \times 100$.

Controls: Take standards through neutralisation (and 2-VP) to check recovery.

P14. Ascorbate (ASC/DHA)

Practical reference Noctor, G., & Mhamdi, A. (2022). Quantitative Measurement of Ascorbate and Glutathione by Spectrophotometry. *Methods in Molecular Biology*, 2526, 87–96.

https://link.springer.com/content/pdf/10.1007/978-1-0716-2469-2_6.pdf

Use: Ascorbate pool size and % reduced.

Reagents

- The neutralised HCl extract from P13; 0.1 M phosphate pH 5.6; ascorbate oxidase; DTT; ascorbate standards.

Procedure

1. **ASC:** read A265 of the extract, add ascorbate oxidase, and read again. The ΔA_{265} corresponds to the reduced ascorbate.
2. **Total:** pre-reduce a second aliquot with DTT, then measure as in step 1.
3. Quantify both against ascorbate standards and work quickly on ice.

Calculation DHA = total – ASC; % reduced = ASC / total × 100.

Controls: Take a DHA standard through the DTT step to confirm complete reduction.

Part C – Advanced assays

ID	Assay	Key point	Practical reference
A1	O ₂ • ⁻ release (XTT)	Root tips or segments in XTT; A470; count only the SOD-inhibitable fraction	Frahry & Schopfer 2001 (maize)
A2	RBOH (NADPH oxidase)	Membrane fraction; NADPH-dependent, SOD-inhibitable XTT/NBT reduction; DPI as support only	Sagi & Fluhr 2001
A3	CeCl ₃ cytochemistry (TEM)	Cerium perhydroxide deposits mark H ₂ O ₂ ; resolves apoplast versus symplast	Bestwick et al. 1997
A4	•OH by EPR	4-POBN/ethanol spin trapping; the only rigorous route for •OH	Renew et al. 2005
A5	roGFP2-Orp1 / Grx1-roGFP2	Ratiometric 405/488 nm; DTT/H ₂ O ₂ calibration; no stable cereal lines are published	Nietzel et al. 2019

Part D – QC, resilience metrics, interpretation

QC acceptance (starting points)

Criterion	Accept	If failed
Standard curve	R ² ≥ 0.99; samples inside the range	Dilute and re-run
Technical CV	< 10%	Repeat the sample
Spike recovery (P4)	80–120% per treatment matrix	Correct per matrix
Pooled QC drift	Within ± 15% of the running mean	Check reagents; re-run the plate

Exclude data only for technical failures defined in advance. Randomise plate positions and analyse with mixed models (plate or batch as random effects).

Resilience metrics (time-matched controls)

Metric	Formula
Stress effect	$LRR_t = \ln(X_{\text{stress},t} / X_{\text{control},t})$
Impact (disturbance)	$D = LRR \text{ at peak stress (T2)}$
Recovery	$R = 1 - LRR_{\text{recovery}} / LRR_{\text{peak}} $ (report together with D; unstable when $D \approx 0$)
Composite	Mean of z-scored traits, sign-aligned so that higher = more oxidised

Plot impact against recovery for each genotype (Ingrisch & Bahn 2018).

Interpretation

Pattern	Reading
Transient H ₂ O ₂ + enzyme induction + GSH/ASC return to control + little MDA	Regulated, adaptive response
Persistent H ₂ O ₂ + oxidised GSH/ASC + high MDA	Persistent oxidative imbalance (first check spike recovery and FW/DW)
Strong DAB/NBT staining but unchanged P4 value	Local production, or a change in peroxidase activity or probe uptake
Pool size changes while % reduced is stable	Changed antioxidant capacity rather than oxidation

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